LAB04_IsabellaDias

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Q1. Carregue os pacotes `readxl` e `tidyverse`.

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

Q2. Crie uma variável chamada fname que contenha o caminho para o arquivo `WDIEXCEL.xlsx`, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando `file.exists`

```
fname <- "wdiexcel.xlsx"

file.exists(fname)
```

```
[1] TRUE
```

Q3. Leia a planilha “Country” e armazene no objeto `countries`.

```r
countries <- read_excel(fname, sheet = "Country")
countries
```

```
# A tibble: 263 x 30
   `Country Code` `Short Name`         `Table Name`    `Long Name` `2-alpha code`
   <chr>          <chr>                <chr>           <chr>       <chr>
 1 ABW            Aruba                Aruba           Aruba       AW
 2 AFG            Afghanistan          Afghanistan     Islamic St~ AF
 3 AGO            Angola               Angola          People's R~ AO
 4 ALB            Albania              Albania         Republic o~ AL
 5 AND            Andorra              Andorra         Principali~ AD
 6 ARB            Arab World           Arab World      Arab World  1A
 7 ARE            United Arab Emirates United Arab E~ United Ara~ AE
 8 ARG            Argentina            Argentina       Argentine ~ AR
 9 ARM            Armenia              Armenia         Republic o~ AM
10 ASM            American Samoa       American Samoa American S~ AS
# i 253 more rows
# i 25 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <chr>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

Q4. Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha)

```r
countries <- countries %>%
  filter(`Country Code` %in% c(
    "BRA",
    "AUS",
    "CHL",
    "CAN",
    "JPN"
  ))

countries
```

```
# A tibble: 5 x 30
  `Country Code` `Short Name` `Table Name` `Long Name`             `2-alpha code`
  <chr>          <chr>        <chr>        <chr>                   <chr>
1 AUS            Australia    Australia    Commonwealth of Austr~  AU
```

```
2 BRA            Brazil       Brazil       Federative Republic o~ BR
3 CAN            Canada       Canada       Canada                 CA
4 CHL            Chile        Chile        Republic of Chile      CL
5 JPN            Japan        Japan        Japan                  JP
# i 25 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <chr>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <chr>,
#   `PPP survey year` <chr>, `Balance of Payments Manual in use` <chr>, ...
```

Q5. Leia a planilha "Data" e armazene no objeto WDI.

```
WDI <- read_excel(fname, sheet = "Data")
WDI
```

```
# A tibble: 422,136 x 63
   `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
   <chr>          <chr>          <chr>            <chr>             <dbl>  <dbl>
 1 Arab World     ARB            2005 PPP conver~ PA.NUS.PPP.05        NA     NA
 2 Arab World     ARB            2005 PPP conver~ PA.NUS.PRVT.PP.~     NA     NA
 3 Arab World     ARB            Access to clean~ EG.CFT.ACCS.ZS       NA     NA
 4 Arab World     ARB            Access to elect~ EG.ELC.ACCS.ZS       NA     NA
 5 Arab World     ARB            Access to elect~ EG.ELC.ACCS.RU.~     NA     NA
 6 Arab World     ARB            Access to elect~ EG.ELC.ACCS.UR.~     NA     NA
 7 Arab World     ARB            Account ownersh~ FX.OWN.TOTL.ZS       NA     NA
 8 Arab World     ARB            Account ownersh~ FX.OWN.TOTL.FE.~     NA     NA
 9 Arab World     ARB            Account ownersh~ FX.OWN.TOTL.MA.~     NA     NA
10 Arab World     ARB            Account ownersh~ FX.OWN.TOTL.OL.~     NA     NA
# i 422,126 more rows
# i 57 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>, ...
```

Q6. Selecione apenas as colunas:

- Country Code
- Indicator Code

- E todos os anos de 1960-2018

```
WDI <- WDI %>%
  select(
    "Country Code",
    "Indicator Code",
    "1960":"2018"
  )
WDI
```

```
# A tibble: 422,136 x 61
   `Country Code` `Indicator Code`  `1960` `1961` `1962` `1963` `1964` `1965`
   <chr>          <chr>              <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
 1 ARB            PA.NUS.PPP.05         NA     NA     NA     NA     NA     NA
 2 ARB            PA.NUS.PRVT.PP.05     NA     NA     NA     NA     NA     NA
 3 ARB            EG.CFT.ACCS.ZS        NA     NA     NA     NA     NA     NA
 4 ARB            EG.ELC.ACCS.ZS        NA     NA     NA     NA     NA     NA
 5 ARB            EG.ELC.ACCS.RU.ZS     NA     NA     NA     NA     NA     NA
 6 ARB            EG.ELC.ACCS.UR.ZS     NA     NA     NA     NA     NA     NA
 7 ARB            FX.OWN.TOTL.ZS        NA     NA     NA     NA     NA     NA
 8 ARB            FX.OWN.TOTL.FE.ZS     NA     NA     NA     NA     NA     NA
 9 ARB            FX.OWN.TOTL.MA.ZS     NA     NA     NA     NA     NA     NA
10 ARB            FX.OWN.TOTL.OL.ZS     NA     NA     NA     NA     NA     NA
# i 422,126 more rows
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, ...
```

Q7. Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

```
WDI_tidy <- WDI %>%
  pivot_longer(
    cols = "1960":"2018",
    names_to = "YEAR",
    values_to = "VALUE"
  )
WDI_tidy
```

```
# A tibble: 24,906,024 x 4
   `Country Code` `Indicator Code` YEAR  VALUE
   <chr>          <chr>            <chr> <dbl>
 1 ARB            PA.NUS.PPP.05    1960     NA
 2 ARB            PA.NUS.PPP.05    1961     NA
 3 ARB            PA.NUS.PPP.05    1962     NA
 4 ARB            PA.NUS.PPP.05    1963     NA
 5 ARB            PA.NUS.PPP.05    1964     NA
 6 ARB            PA.NUS.PPP.05    1965     NA
 7 ARB            PA.NUS.PPP.05    1966     NA
 8 ARB            PA.NUS.PPP.05    1967     NA
 9 ARB            PA.NUS.PPP.05    1968     NA
10 ARB            PA.NUS.PPP.05    1969     NA
# i 24,906,014 more rows
```

Q8. Confirme se a coluna referente ao ano está em formato inteiro.

```r
WDI_tidy <- WDI_tidy %>%
  mutate(YEAR = as.integer(YEAR))

str(WDI_tidy$YEAR)
```

```
 int [1:24906024] 1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 ...
```

Q9. Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas: - Country Code - Short Name - YEAR - SP.DYN.CBRT.IN

```r
WDI_tidy <- WDI_tidy %>%
  left_join(
    countries %>%
      select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) %>%
  filter(
    !is.na(`Short Name`),
    `Indicator Code` == "SP.DYN.CBRT.IN"
  ) %>%
  select(
    `Country Code`,
    `Short Name`,
    YEAR,
```

```
    VALUE
  )

WDI_tidy
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`  YEAR VALUE
   <chr>          <chr>        <int> <dbl>
 1 AUS            Australia     1960  22.4
 2 AUS            Australia     1961  22.9
 3 AUS            Australia     1962  22.1
 4 AUS            Australia     1963  21.5
 5 AUS            Australia     1964  20.5
 6 AUS            Australia     1965  19.6
 7 AUS            Australia     1966  19.8
 8 AUS            Australia     1967  19.4
 9 AUS            Australia     1968  20.1
10 AUS            Australia     1969  20.4
# i 285 more rows
```

Q10. Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

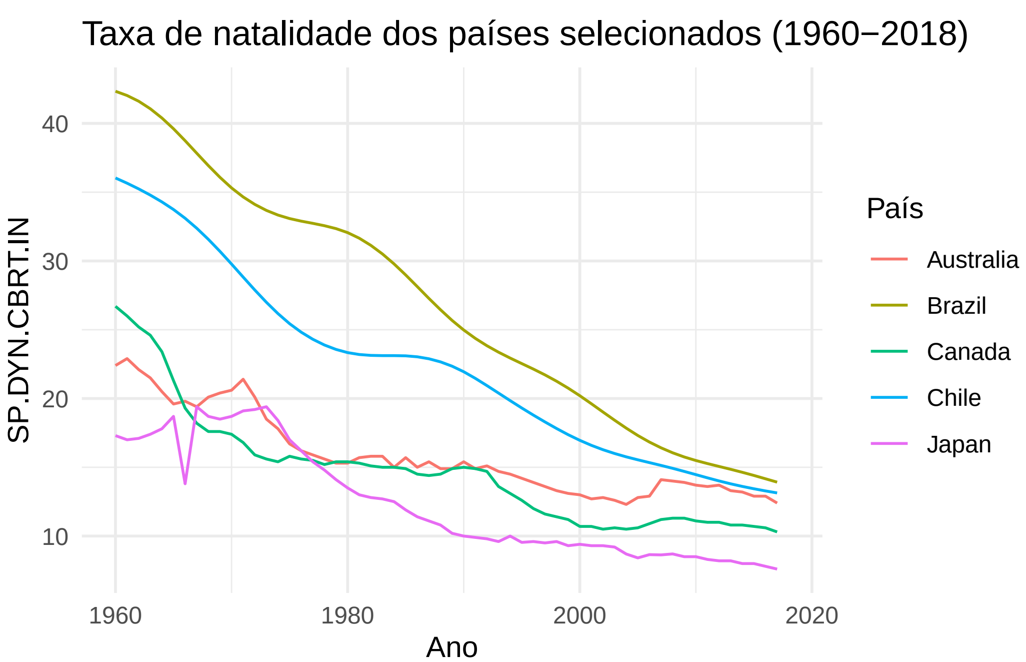
```
WDI_tidy <- WDI_tidy %>%
  arrange(`Short Name`, YEAR)

WDI_tidy
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`  YEAR VALUE
   <chr>          <chr>        <int> <dbl>
 1 AUS            Australia     1960  22.4
 2 AUS            Australia     1961  22.9
 3 AUS            Australia     1962  22.1
 4 AUS            Australia     1963  21.5
 5 AUS            Australia     1964  20.5
 6 AUS            Australia     1965  19.6
 7 AUS            Australia     1966  19.8
 8 AUS            Australia     1967  19.4
 9 AUS            Australia     1968  20.1
10 AUS            Australia     1969  20.4
# i 285 more rows
```

Q11. Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```
ggplot(WDI_tidy, aes(x = YEAR, y = VALUE, color = `Short Name`)) +
  geom_line() +
  labs(
    title = "Taxa de natalidade dos países selecionados (1960-2018)",
    x = "Ano",
    y = "SP.DYN.CBRT.IN",
    color = "País"
  ) +
  theme_minimal()
```

```
Warning: Removed 5 rows containing missing values or values outside the scale range
(`geom_line()`).
```



Horário

```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-09 08:41:04 -03"
```